Qingyang Liu

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Personal Website: https://xxxco.github.io

Education

Duke University, School of Medicine, Durham, NC *Aug. 2024 – May 2026*
Master of Biostatistics

University of Washington, Seattle, WA *Sept. 2020 – June 2024*
BS in Applied Computational Mathematical Science

Experience

Machine Learning Research Assistant, Duke University Rohit Singh Lab *Oct. 2024 - Present*
- Built a scalable ML framework across 33 biological networks using PageRank, diffusion models, and node embeddings
- Improved pipeline speed by 50% and ranking accuracy by 30% through algorithmic and data-flow optimizations
- Developed single-cell integration pipelines with GeoSketch pseudobulking and DE-based feature filtering
- Designed modular Python workflows combining structured and unstructured genomic data for ML analyses

Data Science Research Assistant, Duke Center for AIDS Research *Oct. 2024 - Present*
- Ran large-scale clustering and unsupervised ML on genomic datasets to uncover robust population patterns
- Evaluated model quality using TCA, V-measure, and other metrics to improve clustering reliability
- Optimized parameters and detection thresholds to enhance sensitivity and specificity if marker identification
- Delivered insights through region- and subtype-level comparative analyses to support surveillance decisions

Algorithm Engineering Intern, Ping'an Technology *July 2023 – Sept. 2023*
- Applied interpolation and curve fitting to improve synchronization between speech and lip movements in AI figures
- Automated analysis of 80,000+ videos in Python, extracting metadata and streamlining workflows for efficiency
- Built and managed databases by uploading and classifying 11,000+ minutes of video data for AI training pipelines
- Processed and optimized raw video data with FFmpeg, adjusting parameters to maximize clarity and visual quality

Projects

Pilot Trial Designing Software, Duke University *Aug. 2024 - Present*
- Built simulation and Bayesian models to assess pilot-trial feasibility, dropout risk, and data-driven design decisions
- Developed a Shiny app for automated scenario testing and analysis, improving planning efficiency and guidance

LLM Safety & Interpretability with Sparse Autoencoders, Duke University *Nov. 2025 – Dec. 2025*
- Identified toxicity- and hallucination-aligned SAE features and built classifiers with strong detection performance
- Applied feature-level steering to hidden states to reduce unsafe generations and improve answer accuracy

Reinforcement Learning Algorithm Benchmarking, Duke University *Nov. 2025 – Dec. 2025*
- Benchmarked DQN, A2C, and PPO with ablations to assess key algorithm components across control tasks
- Analyzed stability and efficiency, highlighting actor-critic advantages in continuous-control environments

Vaccine Reservation System, University of Washington *Dec. 2021*
- Designed a relational SQL database with Java integration, including queries, procedures, and supply-chain simulation
- Built user-facing modules for secure login, appointment scheduling, and availability monitoring for vaccine reservation

Skills

Programming Language: Python, R, SQL, Java
Machine Learning: PyTorch, Scikit-learn, LightGBM, TensorFlow, XGBoots, Cross-Validation
Data Analysis: A/B Testing, Statistical Modeling, Pandas, NumPy, Matplotlib, Seaborn, Shiny, Data Visualization
Tools & Platforms: Git, Docker, FFmpeg, LaTeX, Linux